Walk calibration of BTL modules in testbeam:

Via Synchronization of MCP scope data with FEBd

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Discussion draft; not approved for public release.

Purpose: provide a reliable time reference for BTL analyses

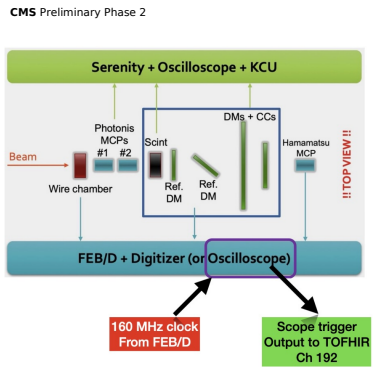
Several BTL test-beam analyses need a reliable particle-arrival time reference.

For a subset of the data, a digital oscilloscope recorded MCP-PMT signals in parallel with the BTL readout.

Approach: synchronize the two streams, use the MCP to calibrate upstream reference bars, then use their corrected times as the reference for downstream timing.

Final check: the time difference between downstream and reference bars before and after walk correction.

Parallel MCP and BTL streams are linked through FEBd

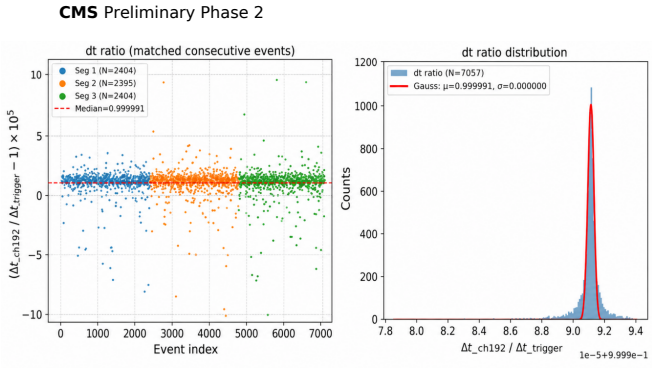


Two readout streams
The scope records the MCP-PMT pulse. FEBd records the BTL SiPM signals.

Synchronization link
The scope trigger also reaches TOFHIR channel 192 in FEBd. This link connects independently clocked records of the same particle.

September 2025 beam test. The highlighted lower path provides the reference-bar data. This FEBd-based synchronization is specific to the test-beam setup, not the final HL-LHC synchronization scheme.

Events are correlated using the intervals between triggers

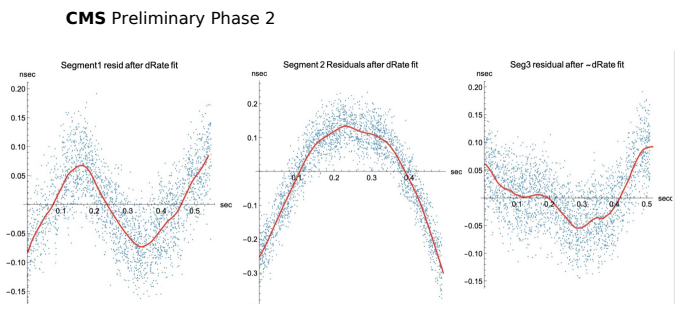


1. Compare consecutive trigger intervals in channel 192 and the scope.
2. Find consistent interval patterns.
3. Keep only unique event matches.

Consistent trigger patterns identify coincident events in the two streams.
Event matching alone does not remove relative clock drift.

The correlated intervals and their ratio test event matching, not detector resolution. GitHub: trigger matching.

The matched clocks show low-frequency relative drift



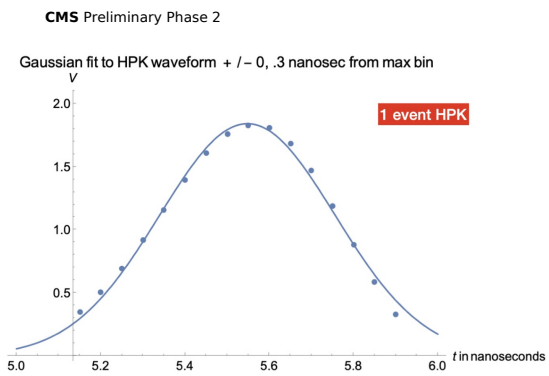
Correct the timing coordinate

1. Split data at clock resets.
2. Fit a line between matched trigger times in each segment.
3. Fit the remaining drift with LOWESS.

Add the drift to the line prediction to obtain $g(u)$, the expected TOFHIR trigger time.

Points show residuals before LOWESS; red curves describe smooth drift. Reject groups with pre-LOWESS residual scale $1.4826\,\mathrm{MAD} > 1$ ns. GitHub: clock reconstruction.

After synchronization, the MCP defines the calibration time



Fit the MCP waveform to estimate its pulse time.
Within the scope waveform, define

$$\delta_{\mathrm{MCP}} = t_{\mathrm{peak}} - t_{\mathrm{trigger}}.$$

For SiPM channel i, use

$$\Delta t_i = t_i - g(u) - \delta_{\mathrm{MCP}}.$$

The charge-dependent part of this difference is the walk.

The waveform and fitted template determine the MCP pulse time. u is the scope-trigger timestamp. A fixed cable/geometry offset is irrelevant to the shape; clock-rate conversion over the short waveform interval is neglected.

Measure the upstream channels' time walk versus charge

Measure the charge-dependent threshold delay relative to the synchronized MCP.

1. In run 4399, require exactly five TOFHIR hits and a unique MCP match: the target upstream pair, downstream channels 234 and 243, and channel 192.
2. Require waveform and clock quality, finite positive charge and slope-quality quantities. Keep charge at or above the spectrum's left half-maximum boundary.
3. Form 20 uniform $1/Q$ bins. Clip timing outliers at $5(1.4826\,\mathrm{MAD})$ from each bin's median; require at least five entries.
4. Fit the bin means with weights $1/\mathrm{SEM}^2$:

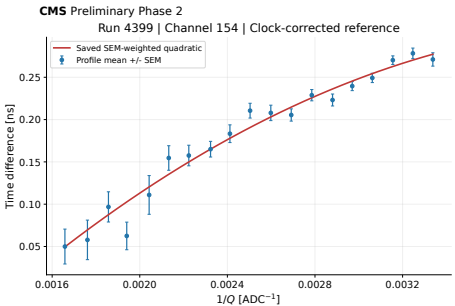
$$W_i(Q) = a_i + b_i/Q + c_i/Q^2.$$

Q is the measured ADC charge proxy, not energy in MeV. SEM denotes the standard error of the mean.

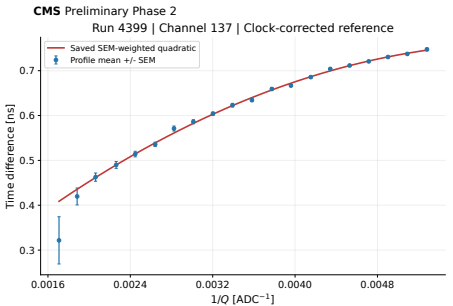
Reference pairs: bar 7 = channels 133 and 154; bar 8 = channels 137 and 150. GitHub: selected reference sample.

Walk-corrected upstream bars provide the time reference

Bar 7: channel 154



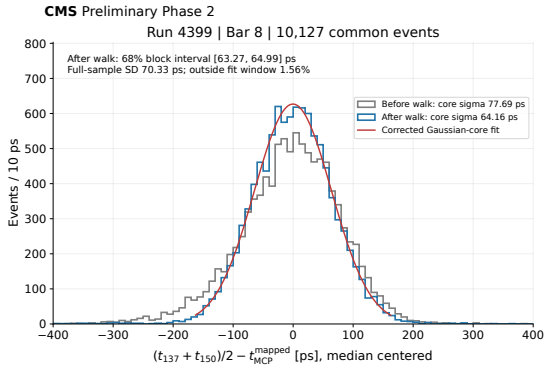
Bar 8: channel 137



Correct each end: $t_i^{\mathrm{corr}} = t_i - W_i(Q_i)$. Use the bar average: $t_{\mathrm{ref}} = (t_L^{\mathrm{corr}} + t_R^{\mathrm{corr}})/2$.

Run 4399 examples. Points show mean time differences with SEM; red curves are SEM-weighted quadratics in $1/Q$. Only populated charge support is shown. The $+26$ ns display offset does not change the walk shape. The full note also shows channels 133 and 150.

Check the calibrated reference bar against the MCP



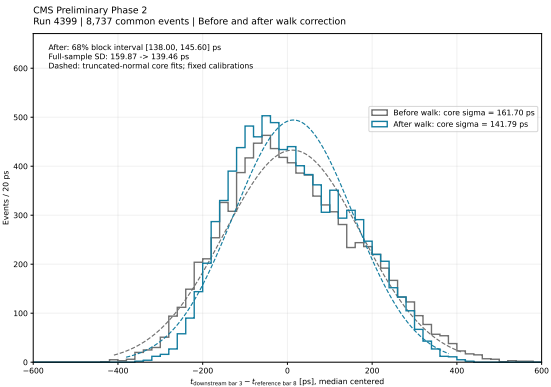
10127 common events
Subtract each channel's fitted walk, then form the bar-average time minus MCP time.
Core width

$$77.69 \longrightarrow 64.16 \text{ ps}$$

Corrected full-sample SD: 70.33 ps.
This is not the intrinsic bar resolution.

Fixed clock mapping; a truncated-normal core fit uses median $\pm 2.5(1.4826\,\mathrm{MAD})$. The corrected-width 16–84% interval is [63.27, 64.99] ps from 400 file/segment-block bootstraps with fixed calibrations.

Final check: downstream–reference-bar timing



8,737 common events
Bar 3 (234, 243) minus reference bar 8 (137, 150).
All four channels are walk corrected before averaging each bar's ends.

Core width

$$161.7 \longrightarrow 141.8 \text{ ps}$$

12.3% narrower. The MCP time cancels from this difference.

Same events and populated charge support; histograms median-centered for display. Same core-fit definition as the preceding slide; fixed-calibration 68% block interval: [138.0, 145.6] ps after correction. In-sample, asymmetric and fit-window dependent; not a single-bar resolution measurement.

References

[1] GitHub repository, `https://github.com/De-Cristo/LecroyTranslator4CMSTimingDetector`.

[2] Presentation by the Saclay group. `https://indico.cern.ch/event/1619706/`

[3] "Timing Performance of a Micro-Channel-Plate Photomultiplier Tube", `https://arxiv.org/pdf/1909.12604`

[4] CMS Collaboration, "A MIP Timing Detector for the CMS Phase-2 Upgrade". Technical Report CERN-LHCC-2019-003. CMS-TDR-020, CERN, Geneva, 2019 and "A High-Granularity Timing Detector for the ATLAS Phase-II Upgrade", CERN-LHCC-2020-007; ATLAS-TDR-031

[5] S. White, Nov. 2025 BTL TB Analysis mtg presentation, `https://indico.cern.ch/event/1614521/contributions/6803723/attachments/3178820/5922530/5vs10vs20GSa_s.pdf`

[6] W. R. Leo, "Techniques For Nuclear and Particle Physics Experiments", p.326, Springer-Verlag Berlin Heidelberg.

[7] S. White and A. Boletti, "Amplitude Walk in Fast Timing: The Role of Dual Thresholds" 2026 JINST V21 P02001.

[8] W. S. Cleveland, "Robust Locally Weighted Regression and Regression Scatterplots", J. Amer. Statist. Assoc. **74** (1979) 829.

Method for Walk Correction using Leading Edge Slope:

Derived from dual threshold feature of TOFHIR2C

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Discussion draft; not approved for public release.

Abstract

Two discriminator thresholds measure a finite-interval slope on the pulse's leading edge. A linear fit describes the mean slope as a function of charge. Its inverse provides a time-walk model with one amplitude, AWC, per channel.

This study measures slopes in run 4076 and determines AWC by comparison with independent MCP calibration curves from run 4399. The fit uses only charge intervals populated in both runs.

The comparison tests calibration shapes, not corrected-event timing resolution. The present AWC calibration still needs the MCP reference; reference-free determination remains a future step.

Two thresholds measure how fast the pulse rises

Problem: a fixed threshold gives an amplitude-dependent timestamp. A steeper leading edge generally crosses the threshold earlier.

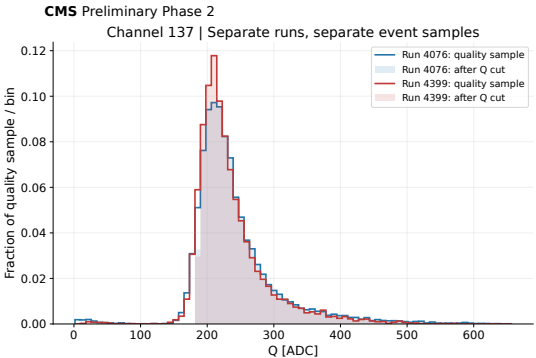
$$I_1 < I_2, \qquad t_1 < t_2, \qquad S = \frac{I_2 - I_1}{t_2 - t_1} \quad [\mu\mathrm{A/ns}].$$

1. Reconstruct both threshold-crossing times with consistent coarse and fine timing.
2. Divide the threshold-current separation by the crossing-time separation.
3. Use charge Q to describe the mean slope for each channel.

S is a finite-interval proxy, not the instantaneous derivative or the current difference itself.

Run 4076 uses $\Delta I = 7.8125\,\mu$A from the implemented threshold conversion. GitHub: timing and threshold reconstruction.

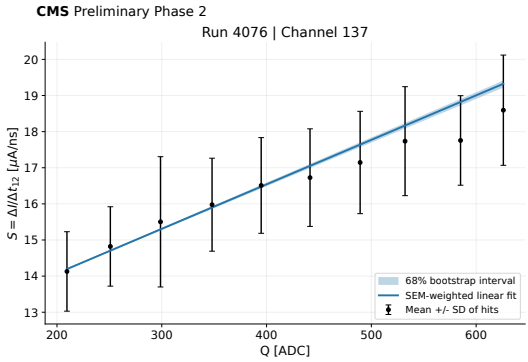
1. Select the prompt population and the charge peak region



1. Require one hit per target-bar end; no other hit in that module.
2. Leave the other module unconstrained; ignore channel 192.
3. Require finite Q, S, positive threshold and time separations, and the same coarse bin.
4. Keep $Q \geq Q_{\mathrm{left}}$.

Channel 137: lines show quality-selected charge spectra; shading adds the charge selection. Q_{left} is the smoothed histogram's left half-height boundary, not a Landau-fit boundary. Run 4076 starts at about 184.9 ADC; no extra upper-charge or 100 ps minimum-rise cut is applied.

2. Fit the mean slope as a function of charge



Group the selected run-4076 hits into ten charge bins.
Fit profile means with SEM weights:

$$\overline{S}(Q) = c_0 + c_1 Q.$$

Use all selected hits, without a training/test split.
Resample hits and refit 1000 times with fixed bin edges.

Channel 137 has 19 126 selected hits. Points show mean slope; bars show the hit standard deviation (SD). SEM is the standard error of the mean. The band is the conditional 68% bootstrap interval of the fitted mean. Linearity remains a working approximation.

3. Convert the fitted slope into a walk correction

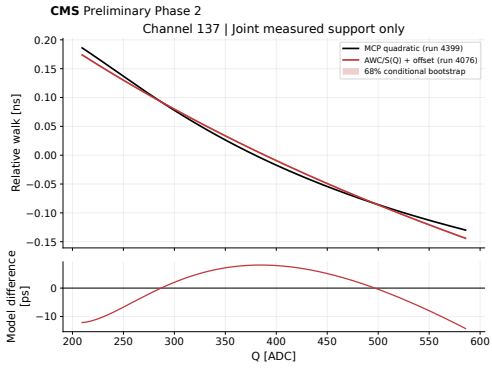
$$W_Q(Q) = \frac{A}{c_0 + c_1 Q}, \qquad t_{\mathrm{corr}} = t_{\mathrm{measured}} - W_Q(Q).$$

- A is AWC, a current amplitude in μA.
- A is constant for one channel and operating configuration.
- The correction changes event by event through Q.

The slope fit determines the charge dependence, but does not determine AWC. The present calibration obtains AWC from an external MCP reference. Use the model only over its calibrated charge range with a positive slope denominator.

The model concerns the mean pulse slope. Its band is not a single-hit timing uncertainty.

4. Determine AWC where both runs have measured charge support



Run 4076: slope model.
Run 4399: MCP quadratic $W_{\mathrm{MCP}} = a + b/Q + c/Q^2$.
Fit A and a free offset B to match

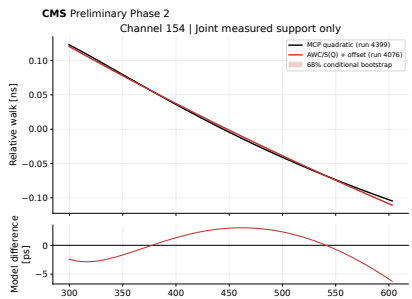
$$\frac{A}{c_0 + c_1 Q} + B$$

to the MCP curve.
Use a uniform charge grid within their populated overlap only.

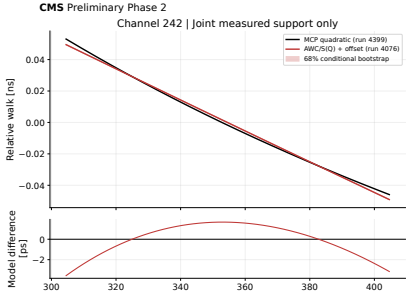
Channel 137: black = MCP reference; red = fitted slope model. The runs contain different events and assume the same nominal setup. AWC = 18.332 μA. The lower panel shows model differences, not event times; RMS mismatch = 6.80 ps.

Apply the same comparison to both detector modules

Upstream: channel 154

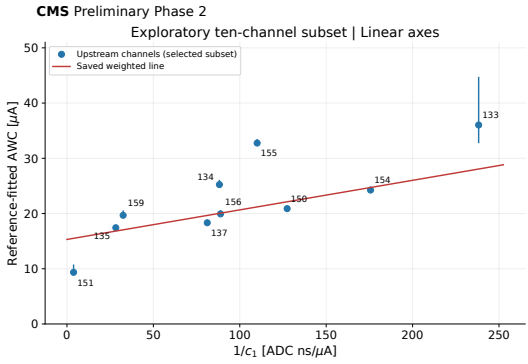


Downstream: channel 242



Each fit uses a separate AWC and offset over the measured overlap. Bands propagate run-4076 slope replicas with the MCP curve fixed. In run 4399, downstream channels require upstream bar 8 in coincidence. Run 4076 leaves the opposite module unconstrained. The opposite module does not supply a timing reference for this fit.

AWC varies between channels beyond a simple slope-coefficient trend



Each point represents one of ten selected upstream channels.
Horizontal coordinate: $1/c_1$.
Vertical coordinate: MCP-calibrated AWC.
The fitted line gives $\chi^2/\mathrm{ndf} = 538.6/8$.
A straight line does not adequately describe this illustrative subset.

Bars are slope-only 16–84% bootstrap intervals. The fit treats x as fixed and uses half the AWC interval as its symmetric error. Reference errors and correlations are omitted. The ten channels were selected for discussion, not by a derived quality rule.

The slope gives the shape; the present AWC still uses MCP

1. Select prompt hits and fit the mean slope versus charge.
2. Form the inverse-slope walk model.
3. Fit AWC and an offset to MCP curves within measured overlap.
4. Apply the charge-dependent correction within the calibrated range.

Uncertainties: displayed bands propagate slope-fit statistics with the MCP curve fixed. They exclude reference-fit, model, selection and cross-run uncertainties.

Next step: validate corrected-event timing and determine AWC from internal information. Shape agreement alone does not demonstrate reference-free calibration.

The full dual-threshold note includes the additional MCP-covariance and charge-weighting checks. GitHub: selection and profiles; GitHub: MCP reference sample.

References

[1] "Performance of the ALICE Time-Of-Flight detector at the LHC", Eur. Phys. J. Plus (2013) 128: 44

[2] "PID performance of the ALICE-TOF detector at Run 2", N. Jacazio in Proceedings of LHCP2018, https://pos.sissa.it/321/232/.

[3] CMS Collaboration, "A MIP Timing Detector for the CMS Phase-2 Upgrade". Technical Report CERN-LHCC-2019-003. CMS-TDR-020, CERN, Geneva, 2019 and "A High-Granularity Timing Detector for the ATLAS Phase-II Upgrade", CERN-LHCC-2020-007; ATLAS-TDR-031

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[6] S. White and Licheng J. Zhang, "September 2025 BTL Testbeam Reference Bar Walk Calibration from Hamamatsu MCP-PMT ",cms note.

[7] E. Albuquerque et al., "TOFHIR2: The readout ASIC of the CMS Barrel MIP Timing Detector", arxiv:2404.01208

[8] J.E. Grund, SLAC-PUB-2416 (Oct. 1979).

[9] A.R. Frolov, T.V. Oslopova, Yu.N. Pestov, "Double threshold discriminator for timing measurements", Nuclear Instruments and Methods Volume 356, Issues 2/3 (1995) https://doi.org/10.1016/0168-9002(94)01239-3.

[10] This will arise in LGAD that are not thinned to remove the effect (J. Vavra, private communication). See also PICOSEC, "Fast Timing with MicroPattern Gas Detectors", where charge in the electron peak is captured-ignoring the ion tail..

[11] W. S. Cleveland, "Robust Locally Weighted Regression and Regression Scatterplots", J. Amer. Statist. Assoc. **74** (1979) 829.

[12] GitHub repository, https://github.com/De-Cristo/LecroyTranslator4CMSTimingDetector.